Mid Term Exam (Spring Semester)
ECO 101
Time – 3 Hours

Make sure that you write clearly and that all parts are well labelled. Do not combine graphs for two separate parts into one.

Q.1 The market demand and supply for ice-cream are given by - $Q_D = 200 - P$ and $Q_S = 4P$. Use these in order to answer the following questions (50 Marks)

a) Find the equilibrium price and quantity for ice-cream using a well labelled diagram. (5+5)

b) With the change in temperature, the demand for ice-cream falls by half. What will happen to the market demand curve for ice-cream, use a diagram to describe the change in the market demand curve for ice-cream? Write down the new equation for the market demand. Does anything happen to the market supply curve following the change in the temperature? Use the new market demand and supply curve in order to solve for the equilibrium. (5+2.5+2.5)

c) Suppose that the government imposes a tax of t per unit on each quantity of ice-cream being bought, use the initial demand and supply equation ($Q_D = 200 - P$ and $Q_S = 4P$) in order to solve for the market equilibrium following the imposition of the tax. Clearly write down the new equation for demand and supply following the tax amount. Represent the same on a well labelled diagram as well. (5+1+4)

d) Use the result in part c) in order to calculate the deadweight loss from taxation, represent the same on a (separate) well labelled diagram as well. (5+5)

e) Suppose that instead of imposing a tax, the government imposes a price ceiling of 10, what will happen to the market equilibrium as a result of the price ceiling. What will happen to the consumer and producer surplus following the price ceiling? (5+2.5+2.5)

Q. 2 Consumer's preferences over two goods Tea and Coffee are described by $U(T, C) = T + C$. Price of Tea ($P_T$) is 5 while the price of Coffee ($P_C$) is 10. Consumer's income equals 100. Use this information in order to answer the following questions -

a) Clearly write down the consumer's problem, describe the particular preferences and also plot the indifference curves and budget constrain. (5+2+2+1)

b) Solve for the consumer's problem (find the chosen consumption bundle) and also provide explanation for the same. Depict the same using a graph as well. (7+2+1)

c) Suppose that the government imposes a tax of t (t>0) per unit on the quantity of tea being purchased, use above information and write down the new budget constraint and plot the same on graph. (5+5)

d) Solve for the consumer's problem, following the imposition of tax, in terms of t. For what values of t, will the consumer choose to consume – a) positive quantity of tea, and b) shift to coffee? (5+5)

e) Suppose that the price of tea rises to 25, write down the new budget constraint and solve for the consumer's problem (find the chosen consumption bundle). Decompose the total effect of a change in the price of tea into substitution and income effect; define these and explain the same using well labelled graphs. (1+1+4+4) **[This part do not follow from part (c) and (d)]**

$y = mx + c$.